

Volume I.
Seventeenth Section. Algebraic Solution of Equations.
Sections 177–182

§ 177. Simplicity of the alternating group

We have seen earlier (§149) that, if the coefficients of an equation of degree n are taken as independent variables and the field of all rational functions of these coefficients is taken as the rationality domain, then the Galois group of the equation is the symmetric group. In that group there is always a normal divisor of index 2, namely the alternating group; the group of the equation is reduced to it when the square root of the discriminant is adjoined.

For four letters the alternating group, consisting of the identity, eight three-cycles, and three products of two transpositions, has the normal divisor of index 3

$$1, (0, 1)(2, 3), (0, 2)(1, 3), (0, 3)(1, 2).$$

This group has three normal divisors of index 2, for instance $1, (0, 1)(2, 3)$, and from this the identity group is obtained by one further step. The group of the twenty-four permutations of four letters is therefore metacyclic; every method of solving the biquadratic equation rests on this fact (§§160, 161).

We shall now show that, for $n > 4$, the alternating group has no normal divisor at all except the identity group; in the terminology introduced above, it is simple. It follows that the necessary condition found for algebraic solvability cannot be fulfilled for equations of degree higher than four whose group is symmetric or alternating. Such equations, as long as their coefficients remain independent variables, are no longer algebraically solvable.

Let A be the alternating group on the n letters $0, 1, 2, \dots, n-1$, and let Q be a normal divisor of A . The permutations of A are generated by three-cycles. Also, if κ and π are permutations, the transformed permutation $\pi^{-1}\kappa\pi$ is obtained by performing the substitution π in the cycles of κ .

If κ is a three-cycle, say $(0, 1, 2)$, then π can be chosen in A so that $\pi^{-1}\kappa\pi$ becomes any desired three-cycle. In

$$\pi = \begin{pmatrix} 0, 1, 2, 3, \dots, n-1 \\ a_0, a_1, a_2, a_3, \dots, a_{n-1} \end{pmatrix}$$

one chooses the first three letters arbitrarily and, if necessary, interchanges two later letters in order that π belong to A . Thus one obtains either (a_0, a_1, a_2) or (a_0, a_2, a_1) , each a power of the other. Hence, if Q contains one three-cycle, it contains all of them, and so $Q = A$.

It remains to prove that every non-identity normal divisor contains a three-cycle. If $\kappa \in Q$ and $\pi \in A$, then $\pi^{-1}\kappa\pi$ belongs to Q , and therefore so does the commutator

$$\lambda = \kappa^{-1}\pi^{-1}\kappa\pi. \tag{1}$$

One chooses π so that, from a non-identity κ , this expression gives a three-cycle, or else a simpler permutation to which a preceding case applies. In forming λ one may ignore the cycles of κ whose letters are left fixed by π , since those cycles cancel between κ^{-1} and κ .

The possible forms of κ are as follows.

1. If κ contains a cycle with more than three letters, say $(1, 2, 3, \dots, m)$, take $\pi = (1, 2, 3)$. Then

$$\lambda = (1, m, m-1, \dots, 2)(2, 3, 1, 4, \dots, m) = (1, 2, 4),$$

a three-cycle.

2. If κ contains two three-cycles,

$$\kappa = (1, 2, 3)(4, 5, 6),$$

take $\pi = (1, 3, 4)$ and obtain

$$\lambda = (1, 3, 2)(4, 6, 5)(3, 2, 4)(1, 5, 6) = (1, 2, 5, 3, 4),$$

which falls under case 1.

3. If κ contains a three-cycle and a transposition,

$$\kappa = (1, 2, 3)(4, 5),$$

then $\pi = (1, 2, 4)$ gives

$$\lambda = (1, 3, 2)(4, 5)(2, 4, 3)(1, 5) = (1, 2, 5, 3, 4),$$

again case 1.

4. If κ contains three transpositions,

$$\kappa = (1, 2)(3, 4)(5, 6),$$

then, from $\pi = (1, 3, 5)$,

$$\lambda = (1, 2)(3, 4)(5, 6)(3, 2)(5, 4)(1, 6) = (1, 3, 5)(2, 6, 4),$$

which falls under case 2.

5. If κ contains two transpositions and one fixed element,

$$\kappa = (1, 2)(3, 4)(5),$$

take $\pi = (1, 2, 5)$ and get

$$\lambda = (1, 2)(3, 4)(5)(2, 5)(3, 4)(1) = (1, 5, 2).$$

For $n > 4$ this exhausts all cases. For $n = 4$ there remains the single case of two transpositions, and that exceptional case is precisely what makes possible the algebraic solution of the equation of the fourth degree.

It follows further that the symmetric group has no normal divisors except itself, the alternating group, and the identity group. Indeed, if S is symmetric, A alternating, and Q a normal divisor of S , then $Q' = A \cap Q$ is a normal divisor of A , hence either A or 1. In the first case Q is A or S . In the second case Q contains no even permutation except the identity; two distinct non-identity elements of Q cannot then occur, and a single such element would have to remain unchanged under every renaming of the letters. This is possible only for two letters, where 1, (1, 2) is already the whole symmetric group.

§ 178. Non-metacyclic equations over the rational numbers

The theorem just proved shows that an equation of degree $n > 4$ cannot be solved algebraically when its coefficients are independent variables. Still more important is the question whether equations of degree n over the rational numbers exist that are not algebraically solvable; more generally, whether there are integral equations whose group is the symmetric group, and hence have no affect.

Form the Galois resolvent $G(t) = 0$ of degree $\nu(n)$ of a general equation $f(x) = 0$ of degree n with undetermined coefficients a . Then $G(t)$ is an integral function of the variables t and a , and

it is not decomposable into factors that are again rational functions of t and a . If values from a field Ω are substituted for the variables a , and if $G(t)$ then becomes reducible over this field, then $f(x) = 0$ has an affect over the rationality domain Ω . The question is therefore whether one can choose the a rationally so that $G(t)$ remains irreducible over the rational numbers.

Hilbert proved the general theorem that, in an irreducible function of any number of variables, rational numbers can be substituted for any chosen part of the variables so that the result remains irreducible in the remaining variables. Weber uses here only a simpler consequence: for every prime degree there are equations without affect.

We proved in §153, 9 that a transitive permutation group on n letters, if n is prime and the group is not the symmetric group, cannot contain a single transposition. Thus, if the irreducible equation $f(x) = 0$ of prime degree n has an affect, its group is transitive but contains no transposition. If any $n - 2$ of its roots are adjoined to the rationality domain, then only the interchange of the two remaining roots could still remain; but that interchange is a transposition, and it is not present. Hence:

1. If an irreducible equation whose degree n is prime has an affect, then any two of its roots can be expressed rationally in terms of the remaining roots.

Consequently:

2. If the field Ω contains only real numbers, then an irreducible equation of prime degree n with an affect in Ω cannot have two imaginary roots and $n - 2$ real roots.

But there are equations of every degree with real coefficients and exactly two conjugate imaginary roots and $n - 2$ real roots. In

$$f(x) = (x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n)$$

choose α_1, α_2 conjugate imaginary and the remaining α real. The number of real and imaginary roots is unchanged under sufficiently small continuous changes of the coefficients; hence the coefficients may be chosen rationally and arbitrarily close to the prescribed real values. It remains only to secure irreducibility.

For this Weber uses Eisenstein's criterion in the following form. Let p be prime and let $c_0, c_1, \dots, c_n$ be integers such that c_0 is not divisible by p , the $c_1, \dots, c_n$ are divisible by p , but c_n is not divisible by p^2 . Then

$$\varphi(x) = c_0x^n + c_1x^{n-1} + \cdots + c_n$$

is irreducible. For if φ split into two factors with rational coefficients, then, after clearing constants as in §2, it would split into factors with integral coefficients,

$$\varphi(x) = (a_0x^h + a_1x^{h-1} + \cdots + a_h)(\beta_0x^k + \beta_1x^{k-1} + \cdots + \beta_k),$$

with $h, k > 0$. Since $a_h\beta_k = c_n$, one of a_h, β_k is divisible by p and the other is not. Suppose β_k is divisible by p , a_h is not, and let β_v be the last coefficient among the β not divisible by p . Then the coefficient of x^{k-v} in the product is not divisible by p , contradicting the assumption on the coefficients of φ .

Thus, by putting the coefficients of $f(x)$ in the form

$$a_1 = \frac{c_1}{c_0}, \quad a_2 = \frac{c_2}{c_0}, \quad \dots, \quad a_n = \frac{c_n}{c_0},$$

one still has enough freedom in choosing the integers c_i to make the rational fractions a_i arbitrarily close to any prescribed system of values, while retaining irreducibility. Hence:

- A. In every prime degree there are infinitely many equations with rational coefficients and without affect.

§ 179. Solution by real radicals

For cubic equations with real coefficients one has long distinguished the cases of negative and positive discriminant. In the latter case the equation has three real roots, while Cardano's formula requires the extraction of a cube root from an imaginary expression. The attempt to put the roots in real form leads back to a cubic equation of the same kind. This is the *casus irreducibilis*. It is a special case of cyclic equations with real roots, where roots of imaginary quantities - equivalently, angle divisions - occur. We can now prove that these imaginary quantities cannot be avoided.

Let Ω be a real rationality domain and let $g(t) = 0$ be a normal equation in Ω . If one root is real, then all roots are real; otherwise the roots occur in conjugate imaginary pairs. If $g(t)$ is the Galois resolvent of an equation $F(x) = 0$, then the roots of F are rationally expressible by one root of g ; hence g has only real roots precisely when F has only real roots.

If $g(t)$ is reduced by adjoining a root θ of an irreducible equation $\chi = 0$ of prime degree, then by §157 all roots of χ lie in $\Omega(\theta)$. Thus, if θ is real, all roots of χ are real. Therefore:

1. A normal equation with real roots can be reduced only by irreducible equations of prime degree all of whose roots are real.

We now ask whether a normal equation can be reduced by adjoining a real radical $\sqrt[p]{a}$. We may assume p prime. If a is not a p -th power in Ω , then

$$x^p - a = 0 \tag{2}$$

is irreducible. Indeed, let $\alpha = \sqrt[p]{a}$ and let ε be a primitive p -th root of unity. The roots are

$$\alpha, \varepsilon\alpha, \varepsilon^2\alpha, \dots, \varepsilon^{p-1}\alpha. \tag{1}$$

If

$$x^p - a = f_1(x)f_2(x), \tag{3}$$

then, for a factor f_1 of degree ν with $0 < \nu < p$, the product of some ν of these roots would be a quantity b of Ω ; hence

$$a^\nu = b^p. \tag{4}$$

Since $(\nu, p) = 1$, choose integers h, k with $\nu h + pk = 1$. Then

$$a = a^{\nu h + pk} = (b^h a^k)^p,$$

contrary to the assumption.

Now assume Ω real. For odd p , equation (2) has imaginary roots and therefore, by 1, cannot reduce a normal equation with real roots. For $p = 2$, this can happen only when $a > 0$ and the degree of $g(t)$ is even. Hence:

2. A normal equation of odd degree cannot be reduced by adjoining a real radical.

This applies to the *casus irreducibilis* of the cubic equation after the square root of the positive discriminant has been adjoined. The rationality domain remains real, and it remains real after any adjunction of real radicals; the normal equation has odd degree and therefore cannot be reduced.

Accordingly, the irreducible cubic equations over the rational numbers divide into cyclic equations of degree 3 with three real roots, which cannot be solved by real radicals, and equations with group of degree six and positive or negative discriminant. The positive discriminant gives the *casus irreducibilis* and the trisection of an angle; the negative discriminant includes as special cases the pure cubics $x^3 = a$, for instance $x^3 = 2$ in the problem of duplicating the cube.

§ 180. Metacyclic equations of prime degree

The general conditions of §176 are still too unwieldy for direct application. We therefore derive, first for equations of prime degree $n > 2$, the criterion first stated by Galois.

Let $f(x) = 0$ be irreducible of prime degree n , and let P be its group. If f is algebraically solvable, then P is metacyclic: there is a chain

$$P, P_1, P_2, \dots, P_{\mu-1}, 1, \quad (1)$$

in which each term is a normal divisor of the preceding one of prime index. The equation itself cannot become reducible before the last adjunction; after that last adjunction it splits into n linear factors. Hence $P_{\mu-1}$ is a cyclic group of degree n . By numbering the roots suitably, its generator may be written

$$\pi = (0, 1, 2, \dots, n-1).$$

Writing the roots as x_z and setting $x_{z+n} = x_z$, the group $P_{\mu-1}$ consists of the substitutions

$$(z, z+h), \quad h = 0, 1, 2, \dots, n-1. \quad (2)$$

These are special cases of the linear substitution

$$(z, az+b), \quad (3)$$

where $a \not\equiv 0 \pmod{n}$. All such permutations form a group of order $n(n-1)$, the linear group. If

$$\lambda = (z, az+b), \quad \lambda' = (z, a'z+b'),$$

then

$$\lambda\lambda' = (z, aa'z + a'b + b'). \quad (4)$$

If g is a primitive root modulo n and a_0 is fixed, every transitive linear group consists of substitutions

$$\lambda = (z, a_0^h z + b), \quad h = 0, 1, \dots, e-1, \quad b = 0, 1, \dots, n-1, \quad (5)$$

where $e \mid n-1$. Conversely, every system of this form is a group.

The key result is:

I. If a transitive linear group L is a normal divisor of another permutation group P on the same n letters, then P too is a linear group.

For the proof, represent an arbitrary permutation π by a function $\varphi(z)$:

$$\pi = \begin{pmatrix} 0, 1, 2, \dots, n-1 \\ a_0, a_1, a_2, \dots, a_{n-1} \end{pmatrix}, \quad a_z = \varphi(z).$$

By Lagrange interpolation modulo n one may write

$$\varphi(z) \equiv -a_0 z^{n-1} - a_1 (z-1)^{n-1} - \dots - a_{n-1} (z-n+1)^{n-1} \pmod{n}. \quad (6)$$

If $L \triangleleft P$ and L contains the cyclic substitution $(z, z+1)$, then, for a permutation $\pi = (z, \varphi(z))$ in P , a linear substitution $(z, a'z+a)$ must exist such that

$$\varphi(z+1) \equiv a'\varphi(z) + a \pmod{n}. \quad (7)$$

Since φ does not reach degree n , comparison of highest powers gives $a' \equiv 1$, whence

$$\varphi(z+h) \equiv \varphi(z) + ah,$$

and therefore

$$\varphi(z) = az + b. \quad (8)$$

Thus every permutation of P is linear.

The Galois theorem follows:

II. The group of a metacyclic equation of prime degree is linear.

For $P_{\mu-1}$ is cyclic, hence linear; since it is a normal divisor of $P_{\mu-2}$, the latter is linear, and so on up to P .

Conversely:

III. Every irreducible equation of prime degree whose group is linear is metacyclic.

Indeed, every transitive linear group L , unless already cyclic, has a normal divisor L' of prime index that is again transitive and linear. In (5), if p is a prime divisor of $e = pe'$, then

$$L' = \{(z, a_0^{ph}z + b) : h = 0, 1, \dots, e' - 1, b = 0, 1, \dots, n - 1\}$$

is such a group.

The full linear group is generated by

$$s = (z, z + 1), \quad t = (z, gz), \quad (9)$$

and a divisor of the form (5) by

$$s = (z, z + 1), \quad t_0 = (z, a_0z). \quad (10)$$

Taking $a_0 = g^2$ gives Kronecker's half-metacyclic group. The full metacyclic group is not a divisor of the alternating group; the greatest common divisor of the two groups is the half-metacyclic group.

The solvability condition can also be expressed by a resolvent. The full linear group is a divisor of index

$$\nu = 1 \cdot 2 \cdot 3 \cdots (n - 2)$$

in the symmetric group. A function y belonging to it satisfies a resolvent $F(y) = 0$ of degree ν . If one root of this resolvent is rational and is not a multiple root, the group of the equation lies in a conjugate metacyclic group. Therefore:

V. The necessary and sufficient condition for the solvability of $f(x) = 0$ by radicals is that the resolvent $F(y) = 0$ of degree ν , chosen so as to have no multiple roots, have a rational root.

Another form is:

VI. If an irreducible equation of prime degree is metacyclic, then all its roots are rationally expressible in terms of any two of them.

For in a linear group no permutation except the identity fixes two letters. Conversely, if an irreducible equation of prime degree has the property that all roots are rationally expressible in terms of any two of them, then no non-identity permutation of its group fixes two letters. Counting the permutations that fix a specified letter and the cyclic permutations of degree n gives a cyclic group $C = 1, \gamma, \dots, \gamma^{n-1}$ as a normal divisor; by theorem I the whole group is linear. Thus:

VII. An irreducible equation of prime degree in which all roots are rationally expressible in terms of any two of them is metacyclic.

For real coefficients one further obtains:

VIII. An irreducible metacyclic equation of odd prime degree with real coefficients has either all roots real or only one real root.

For two real roots would, by VI, make all roots real. In the latter case the sign of the discriminant is $(-1)^{(n-1)/2}$. Hence:

IX. If $n \equiv 1 \pmod{4}$, the discriminant is always positive; if $n \equiv 3 \pmod{4}$, the sign of the discriminant decides which of the two cases of theorem VIII occurs.

§ 181. Application to the metacyclic equations of the fifth degree

For $n = 5$, the cyclic group has 5 permutations, the half-metacyclic group has 10, and the full metacyclic group has 20. A metacyclic function for the general equation of the fifth degree satisfies a resolvent of degree six; if that equation of degree six has a rational root, then the given equation of degree five is metacyclic. The half-metacyclic functions satisfy a resolvent of degree twelve, which after adjoining $\sqrt{\Delta}$ splits into two factors of degree six.

Modulo 5 the generating substitutions are

$$s = (z, z + 1), \quad t = (z, 2z), \quad t^2 = (z, 4z).$$

They act on the letters 0, 1, 2, 3, 4 by

$$\begin{array}{c|ccccc} & 0 & 1 & 2 & 3 & 4 \\ \hline s & 1 & 2 & 3 & 4 & 0 \\ t & 0 & 2 & 4 & 1 & 3 \\ t^2 & 0 & 4 & 3 & 2 & 1 \end{array} \quad (1)$$

and on the five pairs (0, 1), (1, 2), (2, 3), (3, 4), (4, 0) by

$$\begin{array}{c|ccccc} & (0, 1) & (1, 2) & (2, 3) & (3, 4) & (4, 0) \\ \hline s & (1, 2) & (2, 3) & (3, 4) & (4, 0) & (0, 1) \\ t^2 & (0, 4) & (4, 3) & (3, 2) & (2, 1) & (1, 0) \end{array} \quad (2)$$

and

$$\begin{array}{c|ccccc} & (0, 1) & (1, 2) & (2, 3) & (3, 4) & (4, 0) \\ \hline t & (0, 2) & (2, 4) & (4, 1) & (1, 3) & (3, 0). \end{array} \quad (3)$$

Half-metacyclic functions can be formed in many ways. The simplest is

$$u = x_0x_1 + x_1x_2 + x_2x_3 + x_3x_4 + x_4x_0, \quad (4)$$

which t transforms into

$$u' = x_0x_2 + x_2x_4 + x_4x_1 + x_1x_3 + x_3x_0. \quad (5)$$

If

$$f(x) = x^5 - ax^4 + bx^3 - cx^2 + dx - e = 0, \quad (6)$$

then

$$u + u' = b, \quad (7)$$

and

$$y = u - u' \quad (8)$$

is again half-metacyclic, while y^2 is fully metacyclic. If $\sqrt{\Delta}$ is the product of the ten root differences, then

$$Y = \frac{u - u'}{\sqrt{\Delta}} \quad (9)$$

is also fully metacyclic and is a root of an equation of degree six.

Let M denote the full metacyclic group, N the half-metacyclic group, and A the alternating group. Then

$$A = N + N(1, 2)(3, 4) + Nt(0, 1) + Nt(0, 2) + Nt(0, 3) + Nt(0, 4). \quad (10)$$

This gives six values $u_1, \dots, u_6$ conjugate within A ; the corresponding $u'_1, \dots, u'_6$ are obtained by applying t , or equivalently by collecting in each case the pairs missing from the corresponding u_i . In all cases $u_i + u'_i = b$.

The six quantities $y_i = u_i - u'_i$ are roots of an equation of degree six whose coefficients are rational in the coefficients of f and in $\sqrt{\Delta}$. Since all y_i change sign when the sign of $\sqrt{\Delta}$ is changed, the equation has the form

$$y^6 + a_2y^4 + a_4y^2 + a_6 - \sqrt{\Delta}(a_1y^5 + a_3y^3 + a_5y) = 0. \quad (12)$$

The coefficients a_i are integral functions of the coefficients of f . Degree considerations give $a_1 = a_3 = 0$, and a_5 is a number.

Weber computes the resolvent for the Bring-Jerrard form

$$x^5 + \alpha x + \beta = 0. \quad (13)$$

Then a_2, a_4, a_6 do not depend on β , and

$$a_2 = m_1\alpha, \quad a_4 = m_2\alpha^2, \quad a_6 = m_3\alpha^3, \quad a_5 = m. \quad (14)$$

Putting $\beta = 0$ gives

$$x_0 = 0, x_1 = \sqrt{-\alpha}, x_2 = i\sqrt{-\alpha}, \quad x_3 = -\sqrt{-\alpha}, x_4 = -i\sqrt{-\alpha}, \quad (15)$$

and

$$\sqrt{\Delta} = -16\sqrt{\alpha^5}, \quad \Delta = 256\alpha^5. \quad (16)$$

Thus, for $\beta = 0$,

$$\begin{aligned} & y^6 + a_2y^4 + a_4y^2 + a_6 - a_5\sqrt{\Delta}y \\ &= (y + 2\sqrt{\alpha})^4(y^2 - 8y\sqrt{\alpha} + 20\alpha) \\ &= y^6 - 20\alpha y^4 + 240\alpha^2y^2 + 512\sqrt{\alpha^5}y + 320\alpha^3. \end{aligned} \quad (17)$$

Therefore in general

$$y^6 - 20\alpha y^4 + 240\alpha^2y^2 - 32\sqrt{\Delta}y + 320\alpha^3 = 0. \quad (18)$$

The discriminant is

$$\Delta = 5^5\beta^4 + 2^8\alpha^5. \quad (19)$$

For u the equation becomes simpler:

$$u^6 - 5\alpha u^4 + 15\alpha^2u^2 - \sqrt{\Delta}u + 5\alpha^3 = 0. \quad (20)$$

With $u^2 = v$ one obtains the rational equation of degree six

$$(v^3 - 5\alpha v^2 + 15\alpha^2v + 5\alpha^3)^2 = \Delta v, \quad (21)$$

or, equivalently,

$$(v - \alpha)^4(v^2 + 6\alpha v + 25\alpha^2) = 5^5\beta^4v. \quad (22)$$

Thus, to test an irreducible equation of the form (13) for solvability, one first verifies irreducibility and then asks whether (21) or (22) has a rational root. If α, β are integers, such a rational v must be an integer and must divide $25\alpha^6$. In particular, no equation

$$x^5 + 5x + 5t = 0$$

is metacyclic when t is not divisible by 5.

To find metacyclic equations, put in (22)

$$\beta = \alpha\mu, \quad v = \alpha\lambda.$$

Then

$$\alpha = \frac{5^5\mu^4\lambda}{(\lambda - 1)^4(\lambda^2 + 6\lambda + 25)}, \quad \beta = \frac{5^5\mu^5\lambda}{(\lambda - 1)^4(\lambda^2 + 6\lambda + 25)}. \quad (23)$$

If rational values are assigned to λ and μ , then

$$x^5 + \alpha x + \beta = 0 \quad (24)$$

is algebraically solvable. Conversely, no irreducible equation of the form (24) is algebraically solvable unless α, β are representable in the form (23). For example, $\lambda = -1$, $\mu = 1$, after the substitution $x\xi = 5$, gives the irreducible solvable equation

$$\xi^5 + 5\xi^4 - 5 \cdot 64 = 0.$$

§ 182. The group of the resolvent

The six quantities of the preceding section

$$\begin{aligned} v_1 &= (u_1 - u'_1)^2, & v_2 &= (u_2 - u'_2)^2, & v_3 &= (u_3 - u'_3)^2, \\ v_4 &= (u_4 - u'_4)^2, & v_5 &= (u_5 - u'_5)^2, & v_6 &= (u_6 - u'_6)^2 \end{aligned} \quad (1)$$

are the roots of an equation of degree six $F(v) = 0$ whose coefficients are rationally expressible through the elementary symmetric functions a, b, c, d, e of the x . This equation is a total resolvent of the equation of degree five. Its group has the same order as the group of $f(x) = 0$, namely 120. Since the symmetric group on six letters has order 720, we get:

The symmetric permutation group on six letters has a transitive divisor of index 6.

To describe this group C , Weber examines the effect of the four transpositions $(0, 1)$, $(0, 2)$, $(0, 3)$, $(0, 4)$ of the five roots on the indices of the v . This gives the generating permutations

$$\begin{aligned} (0, 1) : \pi_1 &= (1, 3)(2, 5)(4, 6), \\ (0, 2) : \pi_2 &= (1, 4)(2, 3)(5, 6), \\ (0, 3) : \pi_3 &= (1, 5)(2, 6)(3, 4), \\ (0, 4) : \pi_4 &= (1, 6)(2, 4)(3, 5). \end{aligned}$$

Among the 120 permutations of C one finds, for example,

$$\begin{aligned} \pi_1\pi_2 &= (1, 2, 6)(3, 4, 5), \\ \pi_1\pi_2\pi_3 &= (1, 6, 5, 4), \\ \pi_1\pi_2\pi_3\pi_4 &= (2, 4, 6, 3, 5), \\ (\pi_1\pi_2\pi_3\pi_4)^3 &= (2, 3, 4, 5, 6). \end{aligned}$$

A function

$$v_1v_3 + v_2v_4 + v_5v_6$$

is unchanged by π_1 and by $\pi_1\pi_2$. Applying the powers of the cycle $(2, 3, 4, 5, 6)$ yields the five functions

$$\begin{aligned} w_0 &= v_1v_2 + v_4v_5 + v_3v_6, \\ w_1 &= v_1v_4 + v_2v_6 + v_3v_5, \\ w_2 &= v_1v_6 + v_2v_5 + v_3v_4, \\ w_3 &= v_1v_3 + v_2v_4 + v_5v_6, \\ w_4 &= v_1v_5 + v_2v_3 + v_4v_6. \end{aligned} \quad (2)$$

The permutations $\pi_1, \pi_2, \pi_3, \pi_4$ interchange these functions as the transpositions $(0, 1), (0, 2), (0, 3), (0, 4)$ interchange the indices $0, 1, 2, 3, 4$. Thus the symmetric group on the indices of the w corresponds to the group C . A symmetric function of the five w that is not at the same time a symmetric function of the six v belongs to C . One may take, for example,

$$W = w_0^2 + w_1^2 + w_2^2 + w_3^2 + w_4^2$$

or

$$(\lambda - w_0)(\lambda - w_1)(\lambda - w_2)(\lambda - w_3)(\lambda - w_4).$$

Such a quantity W is a root of an irreducible equation of degree six whose coefficients are symmetric functions of the six quantities v .